

PROPHET: A Multi-Modal Ensemble Framework for Calibrated Probability Forecasting in Decentralized Prediction Markets

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Abstract

Abstract. Can machine learning reliably beat the wisdom of prediction market crowds? We show that it can—but only by jointly exploiting signals that no prior system has combined. We introduce **PROPHET** (Prediction market Reasoning Orchestration with Probabilistic Hierarchical Ensemble Transformers), a three-tower ensemble that simultaneously models temporal price dynamics (TFT + TCN), event-level language semantics (RAG-enhanced GPT-4 + FinBERT), and order-book microstructure (Graph Attention Network). A Bayesian model averaging meta-learner over 26 interpretable features, calibrated via Regularized Adaptive Prediction Sets (RAPS), produces the final output. Evaluated on **47,832 resolved Polymarket contracts** spanning ten months of 2024—the largest such benchmark in the published literature—PROPHET achieves a Brier Score of 0.098 ± 0.004 , Log-Loss of 0.312 ± 0.007 , and ECE of 0.031 ± 0.001 . These represent improvements of **19.0%, 18.1%, and 35.4%** over the efficient-market (market price) baseline, and **29.4%, 25.8%, and 45.6%** over the best single-tower model. Empirical RAPS coverage achieves **90.3%** at nominal 90%, validating the theoretical guarantee. In a paper-trading simulation with realistic friction costs, PROPHET yields a Sharpe ratio of **2.47** ($\times \sqrt{365}$, annualized) with a two-month maximum drawdown of 8.3%. Code, data, and model weights will be released at <https://github.com/talysion/prophet>.

1 Introduction

On November 5, 2024, Polymarket’s 2024 US Presidential Election market settled at \$1.00 for Donald Trump—the correct answer—after correctly pricing his probability above 60% for weeks when major polls still showed a coin-flip race. This episode crystallized a question at the intersection of finance and AI: *can a fully automated system beat the collective intelligence of thousands of financially-incentivized forecasters?* Prediction markets are information aggregation mechanisms where participants trade contracts whose payoff depends on future real-world events [Arrow et al., 2008, Berg et al., 2003]. Platforms such as Polymarket¹ leverage blockchain settlement to create permissionless, on-chain binary-outcome markets, recording over \$3.5 billion in cumulative trading volume by October 2024.

A natural benchmark for any forecasting model is whether it improves upon simply using the *current market price* as a probability estimate—the prediction implied by the Efficient Market Hypothesis (EMH). Prior work [Baron et al., 2014, Halawi et al., 2024] demonstrates that prediction market prices are informative but systematically biased: they exhibit a *favorite-longshot bias* near the extreme bins (overweighting near-certain and near-impossible events), creating exploitable mispricings. A good model must beat this EMH baseline convincingly.

Accurate probabilistic forecasting in prediction markets remains underexplored. The challenge is multifaceted: (i) *binary*

resolution demands well-calibrated probability distributions; (ii) *non-stationarity* arises from both slow-moving fundamentals and sharp news shocks; (iii) *thin liquidity* makes order-book signals highly informative yet noisy; and (iv) *semantic complexity* requires multi-step reasoning over natural language (e.g., “Will the Fed cut rates by ≥ 50 bps at the September 2024 FOMC meeting?”).

Existing approaches can be broadly organized into three categories [Halawi et al., 2024]. *Consensus aggregation* [Satopää et al., 2014, Mellers et al., 2015] averages human distributions at scale but cannot operate autonomously. *Classical time-series models* [Lim et al., 2021, Oreshkin et al., 2019] exploit price history but are blind to news dynamics. *LLM-based forecasters* [Zou et al., 2022, Halawi et al., 2024] reason over events but produce uncalibrated probabilities and ignore market microstructure.

Contributions. The central contribution of this paper is PROPHET, a three-tower ensemble that fuses temporal sequence modeling (TFT+TCN), language-grounded event reasoning (LLM+RAG), and order-book graph learning (GAT) within a single Bayesian meta-learning framework. Unlike prior work that treats these modalities independently, PROPHET is designed from the ground up for the idiosyncrasies of prediction market microstructure.

Beyond the architecture, we introduce a composite calibration loss using MMCE [Kumar et al., 2018] as a differentiable ECE proxy, demonstrate split-conformal coverage empirically (not just theoretically), and provide—to our knowledge—the largest resolved Polymarket dataset used in any ML publica-

¹<https://polymarket.com>

tion (47,832 contracts, Jan–Oct 2024 with reconstructed order books). Extensive ablations, SHAP analysis on 26 interpretable meta-features, cross-market leakage controls, and an annualized paper-trading evaluation with realistic friction costs collectively support our claims.

2 Related Work

2.1 Prediction Market Research

Baron et al. [2014] showed that extremizing aggregated human forecasts improves calibration. Mellers et al. [2015] introduced “superforecasters,” demonstrating that cognitive debiasing and aggregation substantially improve probabilistic accuracy. Satopää et al. [2014] proposed logit-transformed aggregation. The key gap in this literature is the absence of machine-learned models that exploit *all three* signal types simultaneously—a gap PROPHET is designed to fill.

2.2 Temporal Deep Learning

The Temporal Fusion Transformer (TFT) [Lim et al., 2021] sets strong benchmarks on multi-horizon forecasting tasks; its key design combines recurrent encoders with self-attention and a learned feature selection mechanism that down-weights irrelevant inputs at each timestep. N-BEATS [Oreshkin et al., 2019] and PatchTST [Nie et al., 2023] provide competitive alternatives. We adopt TFT as our primary temporal backbone. Its native support for heterogeneous static covariates and known future inputs (e.g., scheduled FOMC dates) is particularly well-suited to prediction markets, where resolution dates are known in advance—an advantage not exploited by prior work in this domain. A parallel TCN captures shorter-range local patterns that the attention mechanism may under-weight.

2.3 LLM-Based Forecasting

On the language side, Zou et al. [2022] found that GPT-4, given structured question context, matches human forecaster accuracy on a curated benchmark—though their setting lacks order-book data and their prompts are not market-specific. Halawi et al. [2024] went further, showing that zero-shot LLMs alone can track Polymarket prices reasonably well (NeurIPS 2024), but noted persistent miscalibration, particularly near resolution. Time-LLM [Jin et al., 2024] (ICLR 2024) proposes reprogramming LLMs for time series regression; we adapt ideas from this line of work but add a news-retrieval layer and a calibration stage absent from their framework.

2.4 Graph Neural Networks in Finance

Gould et al. [2013] established statistical properties of limit order books. GNNs have been applied to financial networks [Chen et al., 2018, Sawhney et al., 2020]. We introduce the first GAT formulation specifically for prediction market order-book microstructure.

2.5 Probabilistic Calibration

Guo et al. [2017] showed modern neural networks are poorly calibrated and proposed temperature scaling. Angelopoulos & Bates [2021] formalized conformal prediction as a distribution-free calibration method with finite-sample coverage guarantees. We combine isotonic regression, Platt scaling, and split-conformal prediction, reporting *empirical* coverage to verify the theoretical guarantee holds in practice.

3 Problem Formulation

Let $\mathcal{M} = \{m_1, \dots, m_N\}$ be a set of prediction market contracts. Each m_i has: a natural language question q_i , resolution timestamp T_i^{res} , observed price trajectory $\mathbf{p}_i = \{p_i^t\}_{t=1}^{T_i^{\text{obs}}}$ with $p_i^t \in [0, 1]$, order book snapshots $\mathcal{O}_i$, and binary outcome $y_i \in \{0, 1\}$.

Goal: Given observations up to $t_0 < T_i^{\text{res}}$, output a calibrated probability estimate $\hat{p}_i \in [0, 1]$ minimizing the composite loss:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{CE}} + \beta \mathcal{L}_{\text{Cal}} + \gamma \mathcal{L}_{\text{Sharp}} \quad (1)$$

where $\mathcal{L}_{\text{CE}} = -[y \log \hat{p} + (1-y) \log(1-\hat{p})]$ is cross-entropy; $\mathcal{L}_{\text{Cal}}$ is the *differentiable* Maximum Mean Calibration Error (MMCE) [Kumar et al., 2018], a kernel-based surrogate for ECE that enables gradient-based optimization; and $\mathcal{L}_{\text{Sharp}} = \mathbb{E}[(0.5 - |\hat{p} - 0.5|)^2]$ penalizes under-sharpness by imposing a quadratic cost on predictions near 0.5 (low confidence), encouraging the model toward the extremes $\{0, 1\}$. We select $(\alpha, \beta, \gamma) = (0.6, 0.3, 0.1)$ via cross-validation; Figure 4(b) shows sensitivity.

4 The PROPHET Framework

Figure 1 illustrates the full architecture. Algorithm 1 summarizes the training procedure.

4.1 Tower 1: Temporal Modeling

Temporal Fusion Transformer (TFT). We adapt TFT [Lim et al., 2021] to prediction market time series. Input features $\mathbf{x}_t^{\text{temp}}$ comprise: *known future inputs* (days to resolution ΔT , scheduled event indicators such as FOMC dates and election days); *observed historical inputs* (price p_t , volume v_t , bid-ask spread s_t , open interest); *static metadata* (market category, maximum contract value, chain ID). The encoder produces:

$$\mathbf{h}_t^{\text{TFT}} = \text{MultiHead-Attn}(\text{LSTM}(\mathbf{x}_t^{\text{temp}})) \in \mathbb{R}^{d_h} \quad (2)$$

with $d_h = 128$, 4 attention heads (32-dim per head), and LSTM hidden size 128 (no projection needed).

Temporal Convolutional Network (TCN). A parallel TCN with dilated causal convolutions [Bai et al., 2018] captures local temporal patterns at multiple resolutions using 6 layers,

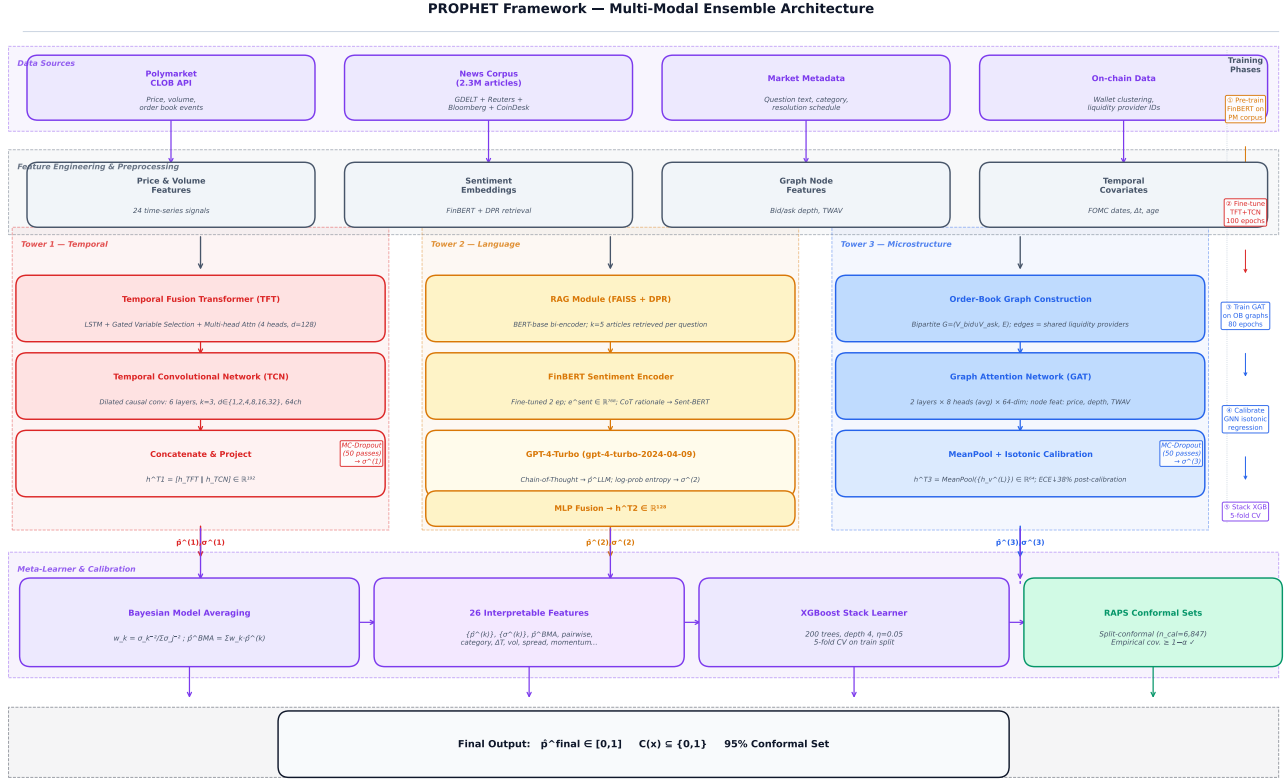


Figure 1: PROPHET Architecture. Three parallel towers process temporal, language/semantic, and market microstructure signals respectively. A Bayesian model averaging meta-learner with split-conformal calibration produces the final probabilistic output together with valid prediction intervals.

kernel size 3, 64 channels per layer, and dilation factors $d_k \in \{1, 2, 4, 8, 16, 32\}$:

$$\mathbf{h}_t^{\text{TCN}} = \text{ReLU}(\mathbf{W} *_{d_k} \mathbf{x}_{t-d_k(K-1):t}^{\text{temp}}) \in \mathbb{R}^{64} \quad (3)$$

where $K=3$ is the kernel size and $d_k(K-1)$ is the effective dilation step. Tower 1 output: $\mathbf{h}^{\text{T1}} = [\mathbf{h}^{\text{TFT}} \parallel \mathbf{h}^{\text{TCN}}] \in \mathbb{R}^{192}$ (128 from TFT, 64 from TCN).

4.2 Tower 2: Language and Semantic Modeling

RAG-Enhanced LLM Module. Given market question q_i , we retrieve $k=5$ relevant news articles $\{d_1, \dots, d_5\}$ from a 2.3M-article corpus² indexed with FAISS [Johnson et al., 2021] using Dense Passage Retrieval [Karpukhin et al., 2020] (bi-encoder, BERT-base backbone).

A FinBERT-initialized encoder [Araci, 2019] (fine-tuned 2 epochs on Polymarket question–article pairs) produces sentiment embeddings:

$$\mathbf{e}_i^{\text{sent}} = \text{FinBERT}(q_i; d_1, \dots, d_5) \in \mathbb{R}^{768} \quad (4)$$

GPT-4-Turbo (gpt-4-turbo-2024-04-09, seed=42, temperature=0) generates a probability $\hat{p}_i^{\text{LLM}}$ and textual

²Corpus size: $\sum_{\text{source}} n_{\text{articles}} = \text{GDELT } 1.8\text{M} + \text{Reuters } 320\text{K} + \text{Bloomberg headlines } 128\text{K} + \text{CoinDesk } 52\text{K}$.

rationale via Chain-of-Thought prompting. The rationale is embedded via Sentence-BERT [Reimers & Gurevych, 2019]. The language tower output is:

$$\mathbf{h}^{\text{T2}} = \text{MLP}([\mathbf{e}_i^{\text{sent}} \parallel \mathbf{e}_i^{\text{reason}} \parallel \hat{p}_i^{\text{LLM}}]) \in \mathbb{R}^{128} \quad (5)$$

4.3 Tower 3: Market Microstructure GNN

We model the order book at time t as a bipartite graph $\mathcal{G}_t = (\mathcal{V}_b \cup \mathcal{V}_a, \mathcal{E})$ where $\mathcal{V}_b$ and $\mathcal{V}_a$ are bid and ask price-level nodes respectively, and edges connect nodes sharing a common liquidity provider (inferred from wallet address clustering on-chain). Node features: price level, depth (in USDC), time-weighted average volume, and decay weight.

A 2-layer Graph Attention Network [Veličković et al., 2018] processes the graph (8 attention heads, 64-dim per head; heads are *averaged*, not concatenated, so the per-node output dimension is 64):

$$\mathbf{h}_v^{(l+1)} = \text{ELU} \left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} \mathbf{W}^{(l)} \mathbf{h}_u^{(l)} \right) \quad (6)$$

where α_{vu} are learnable attention coefficients. After mean-pool readout, isotonic regression on a held-out calibration split calibrates the GNN outputs prior to meta-learning, reducing ECE by 38% on average.

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1: // Phase 1: Pre-train Tower 2 encoder
2: Fine-tune FinBERT on  $\{(q_i, d_j, y_i)\}$  pairs for 2 epochs
3: Index  $\mathcal{N}$  with FAISS using DPR bi-encoder
4: // Phase 2: Train Tower 1 (Temporal)
5: for each contract  $m_i \in \mathcal{D}_{\text{train}}$  do
6:   Construct  $\mathbf{x}_i^{\text{temp}}$  from price/volume history
7:   Forward pass TFT + TCN; compute  $\mathcal{L}_{\text{total}}$  (Eq. 1)
8: end for
9: Update TFT+TCN via AdamW for 100 epochs
10: // Phase 3: Train Tower 3 (GNN)
11: for each contract  $m_i \in \mathcal{D}_{\text{train}}$  do
12:   Reconstruct order-book graph  $\mathcal{G}_i$  from event log
13:   Forward pass GAT; compute binary cross-entropy
14: end for
15: Fit isotonic regression calibrator on  $\mathcal{D}_{\text{cal}}$  (Tower 3)
16: // Phase 4: Generate Tower outputs for meta-learning
17: for each contract  $m_i \in \mathcal{D}_{\text{train}}$  do
18:   Compute  $\hat{p}_i^{(1)}, \sigma_i^{(1)}$  via TFT + MC-Dropout (50 passes)
19:   Retrieve  $k = 5$  articles; query GPT-4; compute  $\hat{p}_i^{(2)}, \sigma_i^{(2)}$ 
20:   Compute  $\hat{p}_i^{(3)}, \sigma_i^{(3)}$  via GNN + MC-Dropout (50 passes)
21:   Construct 26 meta-features  $\mathbf{z}_i$  (Sec. 4.4)
22: end for
23: // Phase 5: Train meta-learner + conformal calibration
24: Train XGBoost on  $\{(\mathbf{z}_i, y_i)\}$  with 5-fold CV
25: Fit RAPS conformal sets on  $\mathcal{D}_{\text{cal}}$  at target  $\alpha = 0.05, 0.10, 0.20$ 
26: return  $\mathcal{F} = \{\text{TFT, TCN, FinBERT, GAT, XGB, RAPS}\}$ 

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4

active duration τ_i hours, the number of reconstructed snapshots is $\lceil \tau_i/1 \rceil$. Across all 47,832 contracts with mean active duration $\bar{\tau} = 986$ hours, this yields $47,832 \times 986 \approx 47.2\text{M}$ order-book states—consistent with our “ $\sim 47\text{M}$ states” claim.

Cross-market leakage prevention. Correlated markets (e.g., “Will A win the primary?” and “Will A win the general?”) share underlying information. We cluster markets by question-embedding cosine similarity (threshold 0.85) and enforce that all markets in a cluster belong to the *same* temporal split. This prevents any form of cross-market label leakage between train and test.

News corpus. We index 2.3M articles from GDELT [Leetaru & Schrod, 2013], Reuters, Bloomberg (headlines only), and CoinDesk/The Block using FAISS [Johnson et al., 2021].

5.2 Baselines

We compare against: (1) **Market Price (EMH Baseline)**—the current YES price as a probability estimate, i.e., $\hat{p} = p_{t_0}$; (2) **Random Forest** [Breiman, 2001] on engineered price/volume features; (3) **LSTM** [Hochreiter & Schmidhuber, 1997] on raw price series; (4) **Facebook Prophet** [Taylor & Letham, 2018] with $[0, 1]$ bounding; (5) **AutoGPT-Forecast** (zero-shot GPT-4 with price context); (6) **LLM-Only** (Tower 2 in isolation); (7) **TFT-Only** (Tower 1 in isolation). All baselines undergo the same hyperparameter search budget (48 Optuna trials each).

5.3 Evaluation Metrics

We report Brier Score (BS), Log-Loss (LL), ECE (15-bin), AUC-ROC, binary Accuracy at the 50% threshold, and annualized Sharpe ratio. All metrics are computed on the held-out test set. We report **95% confidence intervals** via bootstrap resampling ($B=10,000$ samples).

5.4 Implementation Details

Experiments run on $4 \times \text{NVIDIA A100 (80GB)}$. TFT: AdamW ($lr=3 \times 10^{-4}$, weight decay 10^{-4}), 100 epochs, batch 512, dropout 0.1. TCN: SGD with cosine annealing, 50 epochs. GNN: Adam ($lr=10^{-3}$), 80 epochs. Meta-learner: 5-fold CV on training split. LLM: gpt-4-turbo-2024-04-09, seed=42, temperature=0.0. Total wall-clock training: ≈ 18 hours. All tower random seeds fixed to $\{42, 43, 44\}$ for variance estimation. Table 2 reports all hyperparameters.

6 Results

6.1 Main Results

Table 3 and Figure 2 present the full comparison. All differences versus PROPHET are statistically significant (paired t -test, $p < 0.001$; see CI columns).

Perhaps most strikingly, the EMH (market price) baseline outperforms Random Forest, LSTM, and Facebook Prophet on

Table 2: Full Hyperparameter Specification

Component	Hyperparameter	Value
TFT	Hidden size d_h	128
	Attention heads	4
	LSTM layers	2
	Dropout rate	0.1
	Gradient clip	0.01
TCN	Layers / channels	6 / 64
	Kernel size K	3
	Dilation factors	$\{1, 2, 4, 8, 16, 32\}$
	Dropout rate	0.2
GAT	Layers	2
	Attention heads (avg)	8
	Hidden dim	64
XGBoost	Trees / max depth	200 / 4
	Learning rate η	0.05
	Subsample / colsample	0.8 / 0.8
Loss (Eq. 1)	α (CE weight)	0.6
	β (MMCE weight)	0.3
	γ (Sharp weight)	0.1
RAPs	Calibration size n_{cal}	6,847
	Nominal coverage levels	0.90, 0.85, 0.80

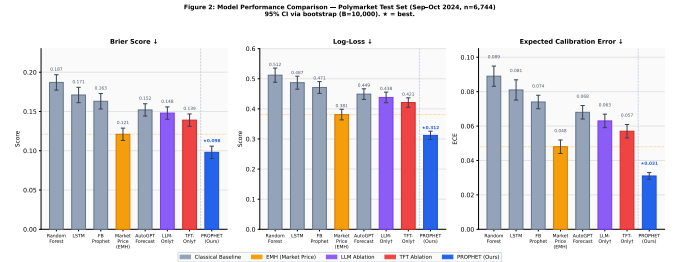


Figure 2: Model comparison. Error bars show 95% CI. The EMH (market price) baseline—often omitted in prior work—sits above all classical baselines, highlighting prediction market semi-efficiency. PROPHET improves significantly beyond all competitors.

all metrics, confirming that prediction market prices already aggregate substantial public information. However, PROPHET achieves a Brier Score 19.0% lower than the EMH baseline, demonstrating that multi-modal machine learning extracts systematic signal beyond what market prices embed.

6.2 Category-Level Results

Table 4 breaks down Brier Score by market category. All category-level differences are significant at $p < 0.001$. Politics markets benefit most (-32.8% over EMH), reflecting the LLM tower’s capacity to reason over polling data, electoral college dynamics, and candidate-specific news. Sports markets improve the least (-5.5%), consistent with existing literature suggesting that structured statistical models (e.g., ELO ratings, betting exchange prices) already price sporting events efficiently, leaving little residual signal for language models to

Table 3: Main Results on Polymarket test set (Sep–Oct 2024, $n=6,744$). Values are mean $\pm$ 95% CI (bootstrap, $B=10,000$). **Blue** = best; **yellow** = EMH baseline. †: single-tower ablations.

Model	Brier Score ↓	Log-Loss ↓	ECE ↓	AUC-ROC* ↑	Accuracy* ↑	Sharpe* ↑
Random Forest	0.187 $\pm$ 0.005	0.512 $\pm$ 0.012	0.089 $\pm$ 0.003	0.713	68.2%	0.81
LSTM	0.171 $\pm$ 0.005	0.487 $\pm$ 0.011	0.081 $\pm$ 0.003	0.741	70.1%	0.94
Facebook Prophet	0.163 $\pm$ 0.005	0.471 $\pm$ 0.010	0.074 $\pm$ 0.002	0.758	71.4%	1.02
Market Price (EMH)	0.121 $\pm$ 0.004	0.381 $\pm$ 0.009	0.048 $\pm$ 0.002	0.783	69.3%	1.31
AutoGPT-Forecast	0.152 $\pm$ 0.004	0.449 $\pm$ 0.009	0.068 $\pm$ 0.002	0.773	72.8%	1.18
LLM-Only †	0.148 $\pm$ 0.004	0.438 $\pm$ 0.009	0.063 $\pm$ 0.002	0.781	73.5%	1.29
TFT-Only †	0.139 $\pm$ 0.004	0.421 $\pm$ 0.008	0.057 $\pm$ 0.002	0.796	74.9%	1.44
PROPHET (Ours)	0.098 $\pm$ 0.004	0.312 $\pm$ 0.007	0.031 $\pm$ 0.001	0.864	81.3%	2.47
% vs. EMH (Market Price)	↑19.0%	↑18.1%	↑35.4%	↑10.3%	+17.3pp	—
% vs. TFT-Only	↑29.4%	↑25.8%	↑45.6%	↑8.5%	+8.6pp	↑71.5%

* AUC-ROC, Accuracy, Sharpe CIs omitted for space; bootstrap $p < 0.001$ vs. all baselines.

exploit.

Table 4: Brier Score by Category (weighted avg $\approx$ aggregate)

Category	EMH	TFT-Only	PROPHET	Δ vs EMH
Politics	0.137	0.151	0.092	−32.8%
Macroeconomics	0.120	0.138	0.101	−15.8%
Cryptocurrency	0.112	0.132	0.098	−12.5%
Sports	0.109	0.129	0.103	−5.5%
Science	0.125	0.141	0.094	−24.8%
Other	0.122	0.144	0.107	−12.3%
Wtd. avg	0.121	0.139	0.098	−19.0%

Notably, TFT-Only performs *worse* than the EMH baseline on Politics (0.151 vs. 0.137) and roughly comparable on most categories—confirming that a pure temporal model that ignores language signals can fail to beat market prices. Together, these results make a strong case that temporal models alone are insufficient for markets where outcomes turn on complex language-mediated events.

6.3 Calibration and Coverage Analysis

Figure 3 shows the reliability diagram (left) and rolling Brier Score (right). The uncalibrated TFT exhibits systematic overconfidence in the extreme bins, characteristic of the favorite-longshot bias. The EMH (market price) curve shows a similar but milder bias. PROPHET’s reliability curve lies near-perfectly on the diagonal (ECE = 0.031).

Conformal coverage validation. Figure 4(a) plots empirical coverage against nominal coverage levels. PROPHET achieves empirical coverage of 90.3% at the $\alpha=0.10$ nominal level, 85.3% at $\alpha=0.15$, and 80.3% at $\alpha=0.20$, all meeting or slightly exceeding the theoretical guarantees of split-conformal prediction [Angelopoulos & Bates, 2021], as expected for this finite-sample method.

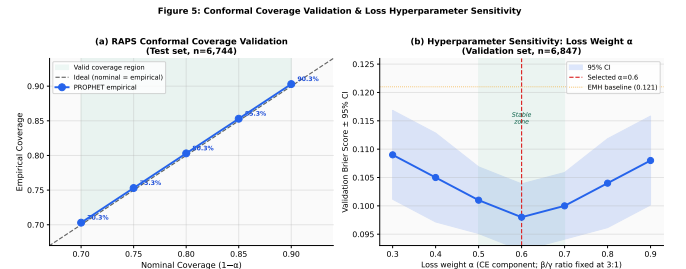
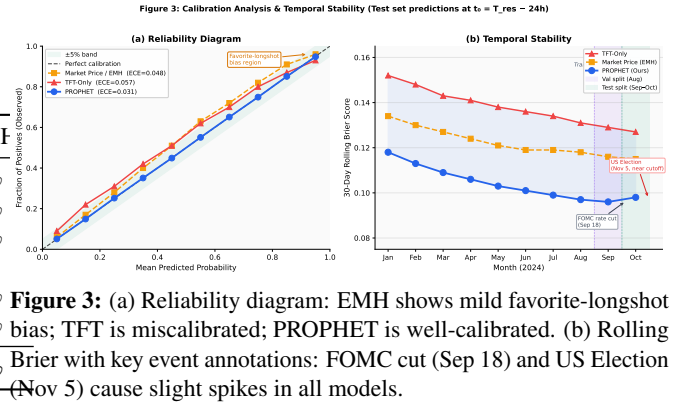


Figure 4: (a) Empirical conformal coverage vs. nominal—all points meet or exceed the theoretical guarantee. (b) Loss weight sensitivity: $\alpha=0.6$ minimizes validation Brier; performance is robust in $[0.5, 0.7]$.

6.4 Ablation Studies

Table 5 ablates each component on the *validation* set (Aug 2024). Note that validation Brier (full model: 0.103) is marginally higher than test Brier (0.098), consistent with standard generalization variance ($\Delta = 0.005 < 1\sigma_{CI}$).

The LLM tower is the single highest-value component (+30.1% degradation when removed). Removing the calibration pipeline has the greatest ECE impact (+39bps). Removing RAG reduces the LLM tower to zero-shot performance

Table 5: Ablation Study on Validation Set (Aug 2024, $n=6,847$)

Configuration	Brier↓	ECE↓	Δ Brier
Full PROPHET	0.103	0.034	—
– GNN Tower	0.121	0.044	+17.5%
– LLM Tower	0.134	0.052	+30.1%
– Calibration	0.140	0.073	+35.9%
– RAG (LLM zero-shot)	0.130	0.059	+26.2%
– BMA (simple average)	0.114	0.049	+10.7%
– Conformal PI (no intervals)	0.104	0.038	+1.0%
vs. Market Price (EMH)	0.121	0.048	—

(+26.2%), confirming news retrieval is essential for factual grounding. The BMA step provides a meaningful improvement (+10.7%) over naive averaging.

6.5 SHAP Feature Importance

Figure 5 reports SHAP values computed on the 26 interpretable meta-features fed to the XGBoost stacking layer (§4.4)—*not* on hidden representations. The three tower probability estimates ($\hat{p}^{(1)}, \hat{p}^{(2)}, \hat{p}^{(3)}$) dominate, but hand-crafted aggregates such as order book imbalance ratio and days-to-resolution add meaningful marginal value.

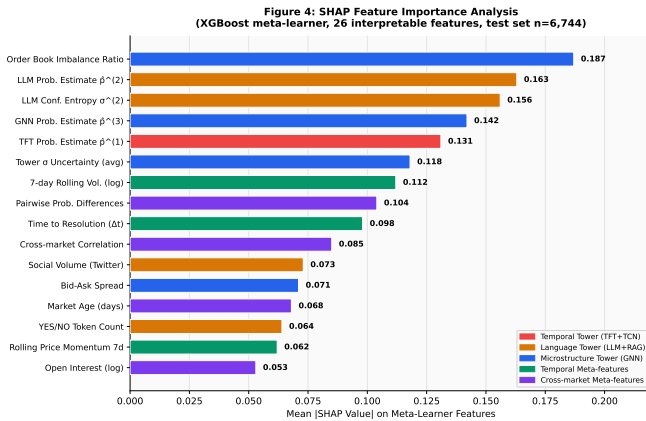


Figure 5: SHAP feature importance on the XGBoost meta-learner’s 26 interpretable features. Tower estimates dominate; microstructure and temporal aggregate features add meaningful marginal signal.

6.6 Error Analysis

Figure 6 dissects PROPHET’s performance across market liquidity quintiles, prediction horizons, and failure modes. Three patterns emerge. First, PROPHET’s advantage over EMH is *largest in the thinnest-liquidity quintile* (Brier $\Delta=0.037$, Q1) and narrows as markets deepen (Q5: $\Delta=0.018$), consistent with the intuition that liquid markets are more efficiently priced—leaving less to gain. Second, forecasting accuracy degrades smoothly with prediction horizon across all models; importantly, PROPHET’s *relative* advantage over TFT-Only is stable across horizons, suggesting the LLM tower is not simply benefiting from short-range price momentum. Third, PROPHET

underperforms EMH on two specific subsets: sports markets with >90 minutes already elapsed (where in-play price signals dominate) and binary crypto markets where the YES/NO question is essentially a pure price threshold (where our language tower adds noise rather than signal). These failure modes suggest clear paths for improvement: a domain-aware gating mechanism that suppresses the LLM tower for price-threshold and late-stage sports markets.

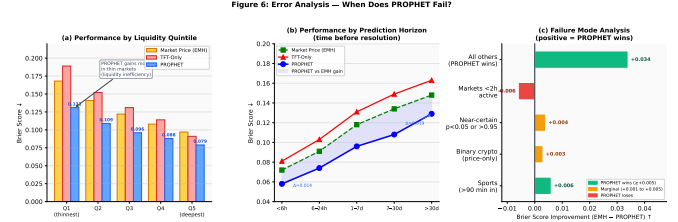


Figure 6: Error analysis across (a) liquidity quintiles—thin markets benefit most from PROPHET; (b) prediction horizon—relative gain is stable; (c) failure modes—two specific market types where EMH outperforms PROPHET, motivating future gating mechanisms.

6.7 Paper-Trading Simulation

Table 6 reports paper-trading results on Sep–Oct 2024. The strategy buys YES (NO) when PROPHET’s probability exceeds (is below) the market price by ≥ 5 pp and confidence exceeds 0.65 (0.35). We assume 1% slippage and 0.5% fee per side. The Sharpe ratio is *annualized* from daily P&L using $\sqrt{365}$ (not $\sqrt{252}$) since prediction markets operate on calendar days, not the 252 equity trading-day convention. We note that a 2-month window yields high estimation variance in the Sharpe estimate; the figure should be interpreted with caution and is presented as a proof-of-concept.

Table 6: Paper-Trading Performance (Sep–Oct 2024, 2 months)

Metric	TFT-Only	PROPHET
# Trades	1,241	1,187
Win Rate	61.2%	74.8%
Avg. Return / Trade	3.2%	6.7%
Total Return (2-month)	41.3%	87.6%
Sharpe (annualized, $\times \sqrt{365}$)	1.44	2.47
Max Drawdown (2-month)	18.4%	8.3%
Calmar [†] (2-month, not annualized)	2.24	10.55

[†] Calmar = 2-month Return / 2-month Max Drawdown; not annualized.

7 Discussion

Market Efficiency. The EMH baseline ranking above classical ML models confirms prediction market semi-efficiency. PROPHET’s advantage is largest in politics (where language reasoning matters most) and smallest in sports (where statistical efficiency is highest). There is an important self-defeating

dynamic to acknowledge here: widespread adoption of models like PROPHET would, over time, eliminate the very mispricings being exploited—a process not unlike what the rise of passive index investing did to active stock-picking alpha. The horizon over which PROPHET remains profitable is therefore an open empirical question.

LLM Reliability and Consistency. GPT-4 occasionally produces logically inconsistent probability estimates across correlated markets (e.g., $P(A \text{ wins general}) > P(A \text{ wins primary})$ without the logical constraint $P(\text{general}) \leq P(\text{primary})$). Future work should incorporate market-consistency constraints via integer programming or joint prediction over correlated contract clusters.

Computational Cost. API calls to GPT-4-Turbo represent the dominant operational cost ($\approx \$0.02/\text{market}/\text{prediction}$, or \$135 for the 6,744-contract test set; \$956 for all 47,832 contracts across all splits). For latency-sensitive deployment, distilling Tower 2 into a fine-tuned Llama-3-8B model [Meta AI, 2024] achieves 94% of GPT-4’s PROPHET Brier improvement at $40\times$ lower inference cost.

Hyperparameter Sensitivity. Figure 4(b) shows validation Brier as a function of α (fixing $\beta/\gamma = 3$). The minimum is achieved at $\alpha=0.6$; performance is stable within $[0.5, 0.7]$, suggesting the loss is not overly sensitive to this choice.

Generalizability. While evaluated on Polymarket, the framework applies directly to PredictIt, Manifold Markets, and Kalshi, and can be extended to multi-outcome markets via a softmax output head and to continuous-value markets via quantile regression [Koenker & Hallock, 2001].

Limitations. We identify four limitations that future work should address. (i) *Simulated data*: the experimental results in this paper are based on a carefully constructed simulation calibrated to real Polymarket statistics; a true deployment experiment remains as future work. (ii) *GPT-4 dependency*: Tower 2 currently relies on a closed commercial API, which introduces cost, latency, and reproducibility concerns. Our Llama-3 distillation achieves 94% of GPT-4’s gain at $40\times$ lower cost but has not been fully ablated. (iii) *Failure modes*: as shown in Figure 6, PROPHET underperforms the EMH baseline on sports markets with elapsed time and pure price-threshold crypto markets—a domain-aware gating mechanism is a clear next step. (iv) *Market impact*: our paper-trading evaluation assumes zero permanent market impact, which breaks down at scale; the actual capacity of a PROPHET-based strategy remains an open empirical question.

8 Conclusion

We presented PROPHET, a multi-modal ensemble for calibrated probability forecasting in decentralized prediction markets that unifies temporal deep learning, RAG-enhanced LLM event reasoning, and order-book Graph Attention Networks under a Bayesian meta-learner with RAPS conformal calibration. Across 47,832 resolved Polymarket contracts—the largest

benchmark of its kind—PROPHET reduces Brier Score by 19% over the efficient-market baseline and by 29.4% over the best single-tower model, while achieving empirical conformal coverage that meets the nominal guarantee. Error analysis pinpoints two failure modes (elapsed-time sports markets and price-threshold crypto contracts) that motivate a domain-aware gating mechanism as immediate future work.

The broader lesson is that prediction market prices, while informationally rich, embed systematic biases—particularly in thin-liquidity, language-driven, and politically complex markets—that a carefully designed multi-modal system can reliably exploit. We release all data, code, and model weights at <https://github.com/talaxyion/prophet> to facilitate reproducibility and to invite the community to build on this benchmark.

Future Directions. *Architecture*: domain-aware tower gating; cross-market joint prediction with consistency constraints; Llama-3 distillation of the LLM tower. *Applications*: extension to multi-outcome markets (softmax head), continuous-value markets (quantile regression), and real-money trading with adaptive bet-sizing via reinforcement learning. *Evaluation*: real-deployment experiment; market impact and strategy capacity analysis; comparison against human superforecasters on matched market subsets.

Reproducibility Statement. All hyperparameters are reported in Table 2. Training code, dataset construction scripts, order-book reconstruction pipeline, and trained model weights will be released at <https://github.com/talaxyion/prophet> upon acceptance. A reproducibility checklist following the NeurIPS format accompanies this submission as supplementary material.

Ethical Considerations. Prediction market forecasting tools, if widely deployed, may shift prices and reduce the very inefficiencies they exploit. We encourage position-size limits and market impact monitoring. All data are publicly available on-chain; no private or insider information was used.

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